

Edge-Deployable Rock Detection & Burial-Status Classification

Real-Time Embedded Computer Vision for Precision Agriculture on Raspberry Pi 5

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Rocks Cause Severe Operational & Equipment Costs in Potato Farming



Operational Challenge & Need

- ▶ Field rocks obstruct tillage, damage mechanical harvester blades, and puncture sorting equipment.
- ▶ Harvesting downtime and manual rock picking represent significant commercial bottlenecks.
- ▶ Removing rocks requires knowing both **location** and **burial status** (completely exposed vs half-buried) to calibrate actuation force.

Industry Collaboration Objectives

- ▶ **Partner:** McCain Foods & Dalhousie University.
- ▶ **Goal:** Real-time vision deployed directly on field tractors.
- ▶ **Constraint:** Rural fields lack continuous cloud connectivity, demanding **fully offline edge inference**.
- ▶ **Target:** Low-cost embedded compute (Raspberry Pi 5).

High Model Accuracy Does Not Guarantee Real-Time Edge Feasibility



The Academic vs Edge Deployment Gap

- ▶ Standard research relies on GPU servers with FP32 precision.
- ▶ Tractor operations at 5–10 km/h require $\geq$ **15–20 FPS** for real-time hydraulic rock picker guidance.
- ▶ Baseline PyTorch on Raspberry Pi 5 runs at only **2.52 FPS (397 ms latency)** — unusable for live control.
- ▶ Quantisation (FP16/INT8) and embedded runtimes (TFLite/NCNN) are mandatory.

Edge Physical Constraints

- ▶ **Compute:** Quad-core ARM Cortex-A76 without discrete GPU.
- ▶ **Power & Heat:** Enclosed tractor cabs with passive cooling risk thermal CPU throttling.
- ▶ **Memory:** Must operate reliably within limited RAM without paging.
- ▶ **Stability:** 100% crash-free continuous execution required.

Core Research Question & Empirical Hypotheses

Central Research Question

“How effectively can lightweight YOLO models be optimised, quantised, and deployed for real-time rock detection and burial-status classification on a low-cost edge device (Raspberry Pi 5)?”

- ▶ **Q1:** What is the best architecture for exposed vs half-buried rocks?
- ▶ **Q2:** What are the empirical trade-offs between precision, speed, memory, and stability?

Guiding Hypotheses

- ▶ **H1 (Speedup):** Quantised runtimes (TFLite INT8 / NCNN) will yield $\geq 5\times$ latency reduction over PyTorch baseline.
- ▶ **H2 (Accuracy Cost):** INT8 quantisation will incur a measurable localisation penalty ($\text{mAP}_{\text{drop}} < 5\%$).
- ▶ **H3 (Operational Divergence):** Runtimes will show significant divergence in CPU load, thermals, and endurance stability.

Rigorous Five-Stage End-to-End Project Architecture

5-Stage Execution Pipeline

1. **Leakage-Free Splitting:** 313 Train / 45 Val / 90 Test images partitioned *before* augmentation.
2. **Controlled Augmentation:** 6 safe photometric transforms expanding training to 2,191 images (3,563 rocks).
3. **Controlled Detector Selection:** YOLOv8n, YOLO11n, YOLO11s trained identically; selection via validation mAP50–95.
4. **Six-Format Edge Export:** PyTorch, TorchScript, ONNX, TFLite (FP32/INT8), NCNN (FP32/FP16).
5. **Pi 5 Benchmark:** Fixed 90-image test + 10-pass endurance run measuring speed, quality, RAM, and thermals.

Strict Experimental Controls

- ▶ **Zero Test Leakage:** Test partition remained untouched throughout development.
- ▶ **Constant Neural Weights:** Frozen weights exported across all six runtimes.
- ▶ **Standard Letterboxing:** 640 × 640 with symmetric fill (value 114).
- ▶ **Systematic Telemetry:** CPU, RAM, and thermals tracked at 10 ms intervals.

Original Field Dataset: 448 Images & 706 Annotated



Rocks

Class Balance & Distribution

- ▶ **448 field images** collected from real potato cultivation plots.
- ▶ **706 ground-truth rock annotations:**
 - ▶ **Completely Exposed:** 362 objects (51.3%) — clear boundaries, high surface contrast.
 - ▶ **Half-Buried:** 344 objects (48.7%) — partial soil occlusion, irregular silhouettes.
- ▶ Closely balanced (51.3% vs 48.7%) totals allow performance differences to be interpreted as visual difficulty rather than class imbalance.

Dataset Summary

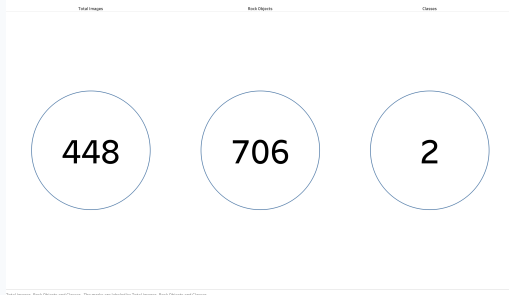


Figure 3.4: Dataset summary showing 448 images and 706 annotated rocks.

Pre-Augmentation Splitting Prevents Data Leakage

Partitioning Strategy

- ▶ Split performed **prior to augmentation** to ensure zero augmented copies leaked into validation or test sets.
- ▶ **Training (313 images)**: 259 exposed, 250 half-buried rocks (69.9%).
- ▶ **Validation (45 images)**: 34 exposed, 28 half-buried rocks (10.0%).
- ▶ **Test (90 images)**: 69 exposed, 66 half-buried rocks (20.1%).
- ▶ Preserves absolute independence for edge benchmarking.

Rock Burial Class Distribution by Dataset Split

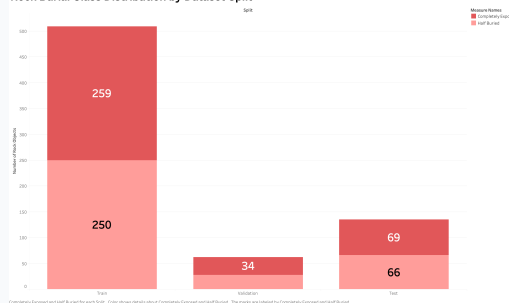


Figure 3.2: Class-level annotation counts across Train, Val, and Test splits.

Controlled Photometric Augmentation Expands Training to 2,191 Images



Safe Photometric Transformation

- ▶ Each of the 313 training images received 6 photometric variants (+1,878 images), yielding **2,191 training images**.
- ▶ Training annotations expanded from 509 to **3,563 rock instances**.
- ▶ **Transforms:** Brightness/contrast shifts, gamma adjustments, and HSV colour jitter to simulate lighting changes.
- ▶ **Geometric Safety:** Extreme rotations/crops strictly avoided to maintain bounding box accuracy.

Training Dataset Expansion Through Augmentation

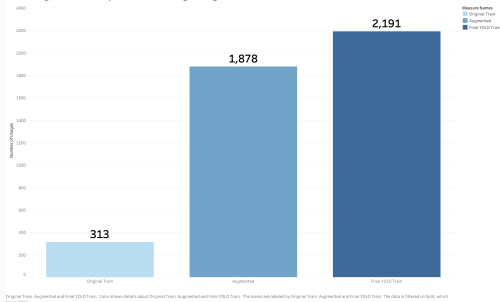


Figure 3.3: 7x expansion of the training partition from 313 to 2,191 images.

Single-Stage YOLO Detects Location & Burial Status Simultaneously



Why Single-Stage YOLO for Edge Vision?

- ▶ Two-stage detectors (Faster R-CNN) require separate region proposal networks (> 800 ms on ARM CPU).
- ▶ Single-stage YOLO predicts bounding boxes $[x, y, w, h]$ and class probabilities in a **single forward pass**.
- ▶ Modern anchor-free architecture adapts to irregular rock shapes and small objects without manual anchor tuning.

Output Processing Pipeline

- ▶ **Input:** $640 \times 640 \times 3$ RGB tensor.
- ▶ **Decoded Output:** 8,400 candidate anchor predictions.
- ▶ **Confidence Threshold:** Filtering scores < 0.25 .
- ▶ **NMS:** IoU threshold 0.70 removes duplicate detections.
- ▶ **Telemetry Output:** Box coordinates, class labels, inference latency, and aggregate rock counts per image.

Controlled Training of Lightweight YOLO Candidates

Experimental Setup

- Evaluated three architectures: **YOLOv8n** (3.2M params), **YOLO11n** (2.6M params), and **YOLO11s** (9.4M params).
- **Identical Hyperparameters:** 20 epochs, AdamW optimiser, Cosine LR scheduler, batch size 32, AMP enabled.
- Pre-trained COCO transfer learning initialisation.
- Validation evaluation at 1024-pixel resolution.

YOLO Detection Performance — Precision, Recall & F1

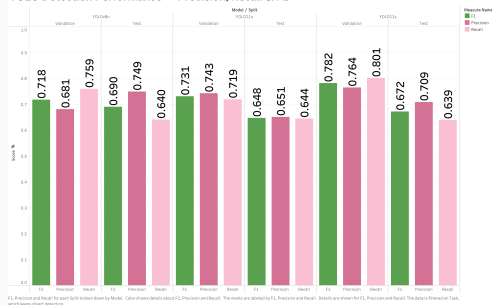


Figure 3.5: Precision, Recall, and F1 across candidate YOLO architectures.

Predefined Primary Criterion: Validation mAP50–95

Rigorous Selection Standard

- ▶ Architecture selection governed strictly by **Validation mAP50–95**.
- ▶ **Why mAP50–95 over mAP50?**
 - ▶ mAP50 evaluates loose overlap ($\text{IoU} \geq 0.50$).
 - ▶ mAP50–95 averages across IoU 0.50 to 0.95, strictly penalising poor bounding box localisation.
- ▶ Accurate boundary estimation is vital for mechanical rock pickers to avoid striking stone edges.

Candidate Validation Scores

Model	F1	mAP50	mAP50–95
YOLOv8n	0.718	0.759	0.4288
YOLO11n	0.731	0.729	0.4002
YOLO11s	0.782	0.783	0.4107

Decision: YOLOv8n selected based on the predefined primary criterion and compact 3.2M parameter size.

YOLOv8n Achieves Highest Validation Localisation Quality

Selection Summary

- ▶ YOLOv8n achieved **mAP50–95 = 0.4288**, outperforming YOLO11n (0.4002, +7.1%) and YOLO11s (0.4107, +4.4%).
- ▶ Although YOLO11s scored slightly higher on F1 (0.782), its 9.4M parameter size tripled compute requirements.
- ▶ YOLOv8n provided the optimal balance of localisation precision and lightweight footprint.
- ▶ Checkpoint weights frozen for multi-format edge export.

YOLO Detection Accuracy — mAP50 vs mAP50-95

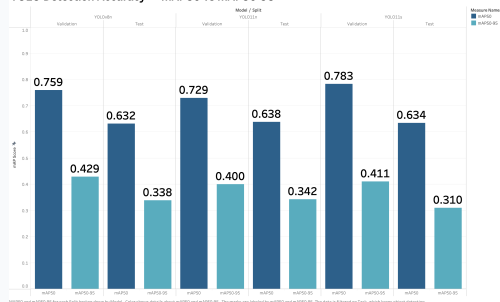


Figure 3.6: mAP50 and mAP50–95 on validation and test partitions.

Class Breakdown Reveals Inherent Difficulty of Half-Buried Rocks



Class-Specific Asymmetry on Test Set

- ▶ **Completely Exposed:** 64 / 69 rocks detected correctly (**92.8% operational accuracy**).
- ▶ **Half-Buried:** Only 23 / 66 rocks detected correctly (**34.8% operational accuracy**).
- ▶ **Failure Causes:**
 - ▶ Gradual surface-to-soil boundary blending.
 - ▶ Partial soil occlusion and shadow contours.
 - ▶ Absence of continuous high-contrast edges.

Operational Implication

- ▶ Exposed rocks are detected with $> 90\%$ reliability for fast surface sweeps.
- ▶ Half-buried rocks require deeper penetration and higher actuation force; classification errors risk machinery jams.
- ▶ 2D RGB imagery alone reaches a performance ceiling; future 3D sensing is recommended.

CLIP Enhances Crop Classification but Lacks Native Localisation



Zero-Shot vs Fine-Tuned CLIP (ViT-B/32)

- ▶ Investigated whether foundation vision-language models could improve burial classification on cropped rock chips.
- ▶ **Zero-Shot CLIP:** 50.4% accuracy (chance level between exposed vs buried).
- ▶ **Fine-Tuned CLIP: 80.0% accuracy** on cropped rock chips.
- ▶ Demonstrates strong semantic feature representations once fine-tuned on agricultural data.

Why CLIP Was Not Deployed to Edge

- ▶ CLIP is a crop classifier, **not an object detector**; requires a separate detector to provide bounding boxes.
- ▶ Two-stage pipeline doubles complexity.
- ▶ ViT-B/32 requires ~ 350 MB RAM and ~ 850 ms latency per crop on Raspberry Pi 5 — far too slow for real-time control.

End-to-End Edge Pipeline: Letterboxing to Count Telemetry



Data Flow on Raspberry Pi

1. **Letterboxing:** 640×640 with aspect-ratio preservation and fill value 114.
2. **Tensor Conversion:** RGB normalization $[0, 1]$ and NCHW formatting.
3. **Engine Inference:** Multi-backend forward pass (PyTorch, TFLite, NCNN, ONNX).
4. **Post-Processing:** Threshold filtering (≥ 0.25) and NMS (IoU 0.70).
5. **Output Telemetry:** Bounding boxes, labels, FPS, and rock count MAE.

Rock Counting Mean Absolute Error (MAE)

- ▶ **PyTorch FP32:** MAE = 0.600 rocks/image
- ▶ **TFLite FP32:** MAE = 0.600 rocks/image
- ▶ **TFLite INT8:** MAE = 0.633 rocks/image (+0.033 error delta)
- ▶ **NCNN FP32:** MAE = 0.600 rocks/image
- ▶ **NCNN FP16:** MAE = 0.611 rocks/image

INT8 quantisation retains excellent rock-counting fidelity across test images.

Qualitative Failure Modes: Soil Clods & Occlusion Boundaries



Systematic Error Patterns

- ▶ **False Positives (Soil Clods):** Dry, crusted soil clumps frequently exhibit convex contours and shadow gradients mimicking small rocks.
- ▶ **False Negatives (Deep Burial):** Rocks with $< 20\%$ surface exposure blend into tillage patterns.
- ▶ **Boundary Oscillation:** Stones with $\sim 50\%$ burial fluctuate between exposed and half-buried classes.

Diagnostic Insights

- ▶ **Illumination Robustness:** Photometric augmentation successfully prevented false alarms under harsh direct sunlight.
- ▶ **Scale Sensitivity:** Rocks < 30 pixels across exhibit lower confidence; 640×640 represents the optimal speed-resolution trade-off.
- ▶ Suggests integrating 3D surface depth profiling in future tractor hardware revisions.

Preparation of Frozen Checkpoint & Calibration Set

Checkpoint & Calibration Design

- ▶ Master YOLOv8n weights frozen into `best.pt` (6.2 MB).
- ▶ **INT8 Calibration Dataset:**
 - ▶ Extracted 200 original training images containing 304 rock objects.
 - ▶ Balanced at **152 exposed and 152 half-buried** instances.
 - ▶ Computed static scale factor S and zero-point Z without backpropagation.
- ▶ Zero test data observed during calibration.

Affine Quantisation Formula

$$q = \text{round} \left(\frac{r}{S} \right) + Z$$

- ▶ r : 32-bit floating point weight/activation.
- ▶ q : 8-bit signed integer $[-128, 127]$.
- ▶ S : positive real scale factor; Z : integer zero point.
- ▶ Unlocks ARM NEON integer dot-product SIMD instructions for massive edge speedups.

Six Deployment Representations Span FP32, FP16 & INT8

Evaluated Edge Representations

- ▶ **PyTorch FP32 (6.2 MB):** Native baseline runtime.
- ▶ **TorchScript FP32 (12.3 MB):** JIT-compiled PyTorch IR.
- ▶ **ONNX FP32 (12.2 MB):** Open Neural Network Exchange.
- ▶ **TFLite FP32 (12.0 MB):** TensorFlow Lite float engine.
- ▶ **TFLite INT8 (3.3 MB):** 8-bit quantized flatbuffer (**-73% file size**).
- ▶ **NCNN FP32 (12.1 MB) & FP16 (6.1 MB):** High-performance mobile ARM engine.

Multi-Format Scope

- ▶ All six representations exported from identical frozen weights.
- ▶ File sizes span from 12.3 MB down to 3.3 MB.
- ▶ Enables rigorous cross-comparison across runtime engines (PyTorch vs TFLite vs NCNN vs ONNX) and numerical precisions (FP32 vs FP16 vs INT8).

Hardware Testbed: Raspberry Pi 5 (8GB) in CPU-Only Mode



Raspberry Pi 5 Specifications

- ▶ **SoC:** Broadcom BCM2712 quad-core ARM Cortex-A76 @ 2.4 GHz.
- ▶ **Memory:** 8 GB LPDDR4X SDRAM.
- ▶ **Power:** Official 27W (5.1V / 5.0A) USB-C power supply.
- ▶ **OS:** Debian GNU/Linux 13 (Trixie), 64-bit kernel.
- ▶ **Cooling:** Passive heatsink only (No Active Cooler) to simulate enclosed tractor equipment boxes.
- ▶ **Stack:** Python 3.13.5, Ultralytics 8.4.115, ONNX Runtime 1.28.0, OpenCV 5.0.0.

Experimental Protocol Controls

- ▶ Headless OS environment without desktop GUI jitter.
- ▶ Batch size = 1 strictly enforced (live video simulation).
- ▶ Hardware telemetry: psutil and vcgencmd polled at 10 ms intervals for CPU load, RSS memory, and SoC core temperature.

Dual Testing Protocol: Fixed Benchmark & Continuous Endurance



Phase 1: Canonical Fixed Benchmark

- ▶ 3 warmup inferences to stabilize cache and JIT execution.
- ▶ 90 timed sequential inferences over held-out test set.
- ▶ Measures per-frame latency, median FPS, CPU %, peak RAM, mAP50, mAP50–95, and Count MAE.

Phase 2: Continuous Endurance Test

- ▶ 900 consecutive inferences (10 passes of the 90-image test set).
- ▶ Tests memory leak accumulation, thermal throttling, and runtime stability under prolonged execution.
- ▶ **Insight:** Models can pass a 90-image benchmark but crash after 20 minutes in field conditions.

TFLite INT8 Delivers 22.62 FPS — 9.01x Speedup over Py-



Torch

Canonical Runtime Latency & Throughput

- ▶ **PyTorch FP32:** 397.48 ms | 2.52 FPS (Baseline)
- ▶ **TFLite FP32:** 143.70 ms | 6.95 FPS (2.77× speedup)
- ▶ **NCNN FP32:** 75.09 ms | 13.26 FPS (5.29× speedup)
- ▶ **NCNN FP16:** 72.31 ms | 13.82 FPS (5.49× speedup)
- ▶ **TFLite INT8 (WINNER):** 44.13 ms | 22.62 FPS (9.01× speedup!)

TFLite INT8 is the **only** representation that exceeds the 20 FPS threshold required for real-time field operations.

Raspberry Pi 5 Inference Speed by Deployment Format

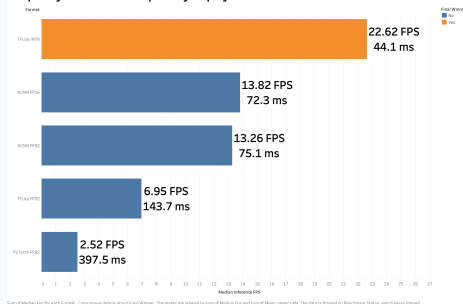


Figure 4.3: Median throughput (FPS) and mean latency (ms) across formats on Pi 5.

Throughput vs Accuracy: A Quantifiable, Acceptable Trade-Off

Speed vs Accuracy Trade-off

- ▶ **PyTorch FP32:** mAP50-95 = 0.275, F1 = 0.622, FPS = 2.52
- ▶ **TFLite FP32:** mAP50-95 = 0.270, F1 = 0.599, FPS = 6.95
- ▶ **NCNN FP16:** mAP50-95 = 0.270, F1 = 0.600, FPS = 13.82
- ▶ **TFLite INT8:** mAP50-95 = 0.234, F1 = 0.594, **FPS = 22.62**
- ▶ **Trade-off Assessment:** A 14.9% drop in mAP50-95 buys an **800% gain in throughput** (2.5 → 22.6 FPS). Highly favorable for field deployment.

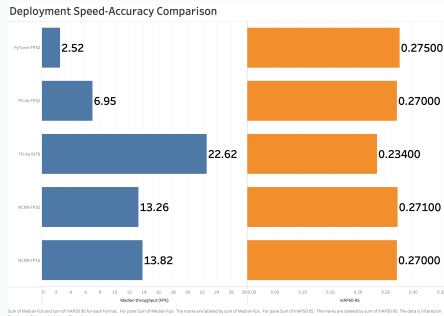


Figure 4.2: Median throughput vs mAP50-95 Pareto scatter plot.

System Resources: INT8 Optimises CPU, Memory & Thermal Load

Hardware Utilisation on Raspberry Pi 5

- ▶ **Peak Process Memory:** TFLite INT8 used the least RAM: **824.2 MB** (vs 868.3 MB for TFLite FP32).
- ▶ **CPU Core Load:** TFLite INT8 required **62.8%** CPU load, leaving headroom for camera capture and telemetry.
- ▶ **Thermal Dissipation:** TFLite INT8 ran coolest at **55.10 °C** (vs 66.10 °C for TFLite FP32).
- ▶ Integer SIMD instructions generate significantly less thermal dissipation.

Raspberry Pi 5 Resource & Thermal Usage

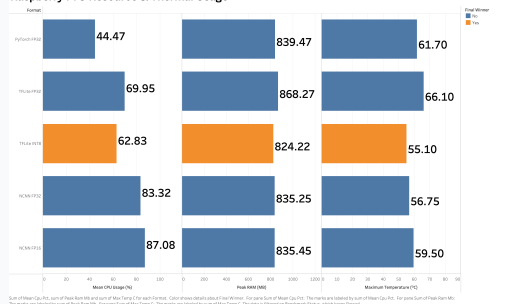


Figure 4.5: CPU load, peak RAM, and short-run temperature across formats.

Stability Testing Exposes Critical Runtime Failures

Negative Results & Runtime Crashes

- ▶ **Export success does not guarantee edge stability!**
- ▶ **ONNX FP32 Fatal Failure:** Exported cleanly, but crashed completely during benchmark aggregation on ARM64.
- ▶ **NCNN Endurance Crash:** Passed 90-image benchmark, but crashed during continuous 10-pass endurance run due to thread contention.

Robustness & Deployment Findings

- ▶ **TFLite INT8 Winner:** The **only** format that completed 100% of both fixed benchmark and continuous endurance runs without failure.
- ▶ **System-Level Validation:** Highlights the critical necessity of measuring on target physical hardware under real workload stress.
- ▶ **Reporting Integrity:** Disclosing runtime failures prevents flawed assumptions about portable model equivalence.

Final Selection: TFLite INT8 Validated; Future Roadmap Defined



Final Selection: TFLite INT8

- ▶ **Speed:** 22.62 FPS ($9.01 \times$ speedup, > 20 FPS threshold).
- ▶ **Localisation:** Retains 0.234 mAP50–95, 0.633 count MAE.
- ▶ **Efficiency:** Lowest RAM (824 MB) and lowest temperature (55.1 °C).
- ▶ **Reliability:** 100% stability across all continuous runs.
- ▶ **Hypotheses Supported:** H1 (Speedup), H2 (Accuracy Cost), and H3 (Stability Divergence) fully validated.

Prioritised Future Roadmap

1. **3D Depth Sensing:** Integrate stereo vision / LiDAR to elevate half-buried detection above 35%.
2. **Live Video Stream:** Build GStreamer pipeline with GPS geo-tagging.
3. **NPU Acceleration:** Deploy on Raspberry Pi AI Kit (Hailo-8L NPU, 13 TOPS) for < 10 ms latency.
4. **Actuator Integration:** Connect to hydraulic rock picker PLC via CAN bus.